

LAMPNODE · CALCULATION NOTE

LampNode sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Radio section rechecked for the US915 band decided on 2026-10-02 (400 ms dwell limit per channel instead of the EU868 duty cycle); SF12 row replaced by SF8; no requirement status changed
0.3	2026-10-01	Amish Chadha	Mass and cost updated for the constructable design (LPN-DDR-003); budget reported as a value-engineering target
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). Firmware port limit (2.5 W above 50 °C inside) added to section 11; R13 now met on paper; engineering changes confirmed
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (energy, self-consumption, switching and inrush, dimming, clock, radar, metering, radio and last gasp, surge, temperature, hosted power, mass, cost) with a status for every requirement

LampNode sizing calculations

On paper, LampNode meets eleven of its sixteen requirements, and none is clearly not met. Four are **at risk**: R3 (relay inrush), R6 (presence detection), R7 (energy saving) and R11 (environment). R13 (hosted sensor power) moved from at risk to met on paper with the firmware port limit decided by Amish on 2026-09-25 (LPN-DDR-002). R1 (socket fit and a 5 min change at height) cannot be verified until a base is tried on a real receptacle.

The largest correction to TRL 2 is the energy saving. Integrating the reference profile over real night lengths at 40° N gives **34.1 to 36.4 %**, not about 41 %, against the 35 % target of R7. The TRL 2 estimate assumed every night had 8 dimmed hours and 3.2 h at full output; in fact the dimmed window 22:00 to 06:00 averages 7.52 h a night, summer nights are shorter than it, and the photocell baseline runs 4,323 h a year, not 4,100 h. The calculations also found three design changes, now in `bom/bom.csv` and decided by Amish on 2026-09-25 (LPN-DDR-002): a temperature-compensated (TCXO) real-time clock, since an ordinary 32 kHz crystal drifts 160 s in a cold month against the 2 min limit of R5; a normally closed relay rated for 80 A inrush and closed at the voltage zero crossing; and a 385 V class surge stage without a gas discharge tube, because the twist-lock socket has no earth contact.

Every number here is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads geometry and part volumes from `cad/src/model.py`, costs from `bom/bom.csv` and the budget from `project.yaml`. All values are first-principles estimates with typical part values; nothing is measured and no part has been chosen from a datasheet.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design.

Input	Value	Basis
Reference luminaire	100 W LED cobra head, 0 to 10 V driver, arm at 7.80 m	LPN-REQ-001 reference case
Location and switching	40° N; lamp on from sunset to sunrise (sun center at -0.833°); clock in local solar time	Mid-latitude city; photocells switch within minutes of sunset
Reference dimming profile	Full to 22:00, 30 % floor from 22:00 to 06:00 with full output 15 % of that time, full from 06:00 to dawn	LPN-REQ-001 assumptions (for estimates only; the road owner sets any profile)
Driver when dimmed	Ideal (input proportional to light), or input = 6 % of rated + 94 % x light level	Typical drivers lose efficiency when dimmed
12 V loads	Controller, clock, accelerometer and light sensor 0.05 W; metering 0.05 W; relay coil 0.40 W by day only; radar 0.20 W and head board 0.03 W by night only	Typical parts
Power supply	5 W module, 0.10 W no-load input, 72 % incremental efficiency at light load, 80 % near full load; full output to 50 °C, falling linearly to 60 % at 70 °C	Typical encapsulated 5 W modules
Hosted sensors	3 W total on two ports, limited to 2.5 W above 50 °C inside; 0.40 W radio transmit transient	LPN-REQ-001 R13, LPN-DDR-002
Driver inrush	60 A cold-start peak at 230 V for a typical 100 W outdoor driver, 47 µF bulk capacitor	Typical driver datasheets; no driver chosen
Radar	24.125 GHz; 8 dBm, 11 dBi each way; half-power elevation half-width 17°; walking person 0.5 m ² ; 35 dB effective noise figure at low Doppler, 10 Hz bins, 15 dB detection threshold	Typical low-cost K-band modules
Tariff	\$0.10 to \$0.25 per kWh	Typical public lighting rates

■ 2. Energy (R7)

At 40° N the lamp is lit **4,323 h a year** (11.84 h a night on average, from 8.99 h at midsummer to 14.68 h at midwinter). The 100 W lamp then uses **432 kWh a year** under a photocell.

Table 2. Annual energy for one 100 W luminaire.

Case	Energy per year	Saving
Photocell, full power	432 kWh	
Schedule only (50 % from 23:00 to 05:00)	324 kWh	25.0 %
Reference presence profile, ideal driver, with 5.9 kWh of controller use	275 kWh	36.4 %
Same, dimmed driver model	285 kWh	34.1 %

The dimmed window covers 2,745 h a year, 7.52 h a night on average. With an ideal driver R7 is met with 1.4 points to spare; with the dimmed driver model it is missed by 0.9 points. **R7 is at risk.** Amish decided on 2026-09-25 to keep the 35 % target and the reference profile and accept this risk until real traffic counts exist (LPN-DDR-002, from

LPN-DDR-001 O2). The TRL 2 method, applied to the same inputs, reproduces its own 236 kWh, so the difference comes from the night lengths and not from an arithmetic slip. The energy saved is **148 kWh a year** with the dimmed driver model.

The presence assumption matters more than the driver. An isolated pedestrian keeps the lamp at full for about 70 s (8.6 s from detection at 15 m to the pole, a 60 s hold and 1 s of ramps), so the 15 % boost equals about 7.8 isolated passes an hour. With random arrivals and the dimmed driver model, the saving is 40.4 % with no traffic, 37.3 % at 4 events an hour, 34.4 % at 8, 29.3 % at 16 and 22.0 % at 30. At 60° N the lamp runs 4,277 h and uses 428 kWh, and the saving falls to 31.8 %, because long winter evenings before 22:00 stay at full output.

■ 3. Self-consumption and supply loading (R10, R13)

The lamp is lit for 0.493 of the year. The relay coil averages 0.203 W, the radar and head board 0.113 W, and the 12 V side 0.416 W in all. The shunt adds 0.8 mW. From the mains LampNode draws **0.68 W** on average, against 1.0 W: R10 is met. The TRL 2 figure (about 0.6 W) used a 0.24 W coil; a typical 16 A relay coil is 0.40 W.

With both hosted ports at 3 W the 12 V side carries 3.50 W by day (coil on) and 3.33 W at night (radar on), or **3.90 W** at peak with a radio transmission: 78 % of the 5 W rating at 25 to 50 °C. Section 11 shows the problem at high temperature and the firmware port limit that solves it.

■ 4. Switching and inrush (R3)

The steady current at 400 W, 100 V and PF 0.90 is 4.44 A (0.93 A for the reference lamp at 120 V), well inside a 16 A contact. The problem is inrush. A typical 100 W outdoor driver takes about 72 A for a few hundred microseconds when switched at a random point of the 277 V wave, and four such drivers on one relay would take about 289 A. Relays made for LED loads are rated for about 80 A inrush, and most are normally open; a normally closed contact with that rating must be found.

Closing at the voltage zero crossing cuts the inrush sharply. The metering IC gives the zero crossing; the relay release time is learned and the coil released early by that amount. With a 0.5 ms timing error the peak falls to about 17 A at 50 Hz and 20 A at 60 Hz for one driver. Contact life is not a concern: two switchings a day for 20 years are 14,600 cycles against 100,000. **R3 is at risk** until a normally closed relay with an 80 A or better inrush rating is found and the partner's driver inrush is known; 400 W from several drivers needs zero-cross closing.

■ 5. Dimming output (R4)

A 12-bit PWM filtered to 0 to 10 V gives 2.44 mV steps against the 100 mV needed for 1 % steps. A 10 mA sink serves 5 drivers at 2 mA of control current, the high end for 0 to 10 V drivers, and dissipates at most 0.10 W. R4 is met by design review for 0 to 10 V; DALI-2 D4i is a later variant (LPN-DDR-001 item 5).

■ 6. Clock without the network (R5)

A 32 kHz tuning-fork crystal (± 20 ppm, falling 0.034 ppm/K² away from 25 °C) drifts 52 s in 30 days at 25 °C, **160 s at -10 °C** and 318 s at -30 °C (230 s at 70 °C). The 2 min limit of R5 is therefore not met in a cold month. A TCXO real-time clock (± 3.5 ppm from -40 to 85 °C) drifts at most **9.1 s** in 30 days. The TCXO clock is now in `bom/bom.csv` line 7 (+\$3), so R5 is met on paper; the change was decided by Amish on 2026-09-25 (LPN-DDR-002). When the network is up, the controller also sets its clock from the LoRaWAN network time.

■ 7. Presence detection (R6)

Geometry. The model puts the radar 7.72 m above the road; tilted 25° down, its beam center reaches the road 16.6 m along the street. A pedestrian (target center 1.0 m up) at 15 m is 24.1° below horizontal and a car or cyclist (0.7 m) at 25 m is 15.7° below, both inside the half-power beam, which covers a walking person from 7.5 to 47.9 m along the street.

Signal. A walking person at 15 m is 16.4 m from the radar in a straight line. With the Table 1 values the echo is -93 dBm against a -129 dBm noise floor: **36 dB** of signal to noise. The 15 dB threshold is reached at 56 m on the beam axis, or 40 m at the beam edge. Range is therefore not limited by signal strength.

Motion and clutter. An approaching pedestrian at 1.4 m/s gives a 205 Hz Doppler tone, a cyclist at 5 m/s 774 Hz and a car at 50 km/h 2,151 Hz. Falling rain gives 136 Hz (drizzle, 2 m/s) to 476 Hz (heavy rain, 7 m/s), which overlaps walkers and cyclists. Because the beam points down, rain always moves away from the radar, while people coming toward the lamp move toward it, so a radar with I/Q (direction) output can ignore receding targets. bom/ bom.csv line 12 now calls for I/Q output. Trees moving in wind produce echoes in both directions and cannot be rejected by direction alone.

Timing. Two 128 ms detection frames, the serial link, the 0 to 10 V output and a 0.3 s driver response give **0.63 s** to full output, inside 1 s. A pedestrian detected at 15 m reaches the pole 10.7 s later.

R6 is at risk. The signal budget and timing are met, but rain, wind-blown trees and the real radar cross-section of a person at this angle can only be settled in the field.

■ 8. Metering (R8)

The current ranges from 66 mA (10 W at 305 V and PF 0.5) to 4.44 A (400 W at 100 V). A 2 mΩ shunt gives 0.131 mV at the low end, 169 times below the metering IC's full scale and inside its 1000:1 range; it dissipates 0.040 W at 4.44 A.

Table 3. Active power error budget (percent of reading).

Term	Error
Metering IC over 1000:1	0.50
Shunt temperature coefficient, 50 ppm/K over 50 K	0.25
Voltage divider temperature coefficient	0.10
Phase error of 0.1° at PF 0.5	0.30
Bench reference used for a one-point calibration	0.50
Worst-case sum (RSS 0.81)	1.65

With a one-point calibration at build the worst case is 1.65 %, inside ±2 %. Without it (1 % shunt and 0.2 % divider tolerances) the worst case is 2.35 %. **R8 is met on paper, provided each unit is calibrated once** against a bench power meter. CalRig, which the TRL 2 note suggested, is a climate chamber for air sensors and cannot do this.

■ 9. Radio, faults and last gasp (R9, R15)

Table 4. Time on air of a 24-byte status message (13 bytes of LoRaWAN overhead, 125 kHz, coding rate 4/5).

Spreading factor (US915 data rate)	Per message	96 per day	Inside 400 ms dwell limit
SF7 (DR3)	82 ms	7.9 s	Yes
SF8 (DR2)	144 ms	13.8 s	Yes
SF9 (DR1)	267 ms	25.7 s	Yes
SF10 (DR0)	494 ms	47.4 s	No , 94 ms over

The first partner is a US partner on the 915 MHz band (decided 2026-10-02), so the radio limits are those of the US915 plan, not the EU868 1 % duty cycle that this note used before. US915 has no duty cycle, but the FCC limits a transmission to 400 ms on one channel. The largest message that fits inside 400 ms is 11 bytes at SF10 (DR0), 53 bytes at SF9 (DR1), 125 bytes at SF8 and 242 bytes at SF7. The 24-byte status message therefore has to go at SF9 (DR1) or faster, where it takes 267 ms and 25.7 s of airtime a day (TRL 2 gave about 0.2 s and 20 s a day before the full 13 bytes of LoRaWAN overhead were counted). A lamp that can only reach its gateway at SF10 (DR0) cannot send the 24-byte status in one message: it would have to send a status of 11 bytes or fewer, or be given a closer gateway. The 6-byte fault message fits at every rate (330 ms at SF10). SF11 and SF12 do not exist at 125 kHz in US915. On a public network with a 30 s daily fair-use limit, lamps at SF10 or slower need a longer interval, as FieldNode found; on a private TwinKit gateway only the dwell limit applies. A 6-byte fault message takes 185 ms at SF9.

For the last message after mains loss, one transmission at SF10 and 22 dBm with two receive windows needs about 0.144 J. A 0.22 F supercapacitor discharged from 5.0 to 3.6 V gives 1.32 J, a margin of 9 times, and recharges in 55 s at 20 mA. R9 is met by design review. R15 is met by design review with the TALQ bridge moved to CityTwin (LPN-DDR-001 item 10).

■ 10. Surge (R11)

The twist-lock socket has no earth contact, so LampNode can clamp only line to neutral; common-mode surges are the luminaire's job. A gas discharge tube to earth, as listed at TRL 2, has nowhere to connect and is dropped. An 8/20 μ s surge carries 23.5 μ s of charge per ampere of peak current. At 5 kA a 385 V class metal oxide varistor clamping at about 1,000 V absorbs **118 J**, within a 20 mm disc's single-surge rating; at 10 kA it absorbs 271 J, which is beyond it. The 320 V class chosen at TRL 2 has only 5 % of margin over the 305 V maximum line voltage; the 385 V class has 26 %. The surge level required by ANSI C136.2 was not read this session and stays to be confirmed. The surge part of R11 is not verifiable without tests.

■ 11. Temperature in the sun (R11, R13)

At 45 °C ambient, with the sun 60° above the horizon, the dome and base present 99 cm² to the sun and absorb 3.46 W at an absorptance of 0.35. Their 338 cm² surface loses heat at 12.4 W/(m² K) in still air. With 0.79 W dissipated inside by day, and the luminaire top 20 K above ambient under the base, the shell rises 13.3 K; allowing 5 K from shell to parts, the inside reaches **63 °C** with the ports unloaded and **65 °C** with 3 W on the ports. That is inside the 70 °C rating, with little margin for a dark dome or a hotter luminaire. **R11 is at risk** (temperature margin, surge and IP66 all unverified).

At 65 °C the derated supply gives about 3.55 W, against the 3.90 W peak with both ports at 3 W, so without a limit R13 would be at risk at the hot end. Amish decided on 2026-09-25 (LPN-DDR-002, from LPN-DDR-001 O3) that the firmware limits the two ports to 2.5 W in total whenever the interior is above 50 °C, rather than fitting a 10 W supply.

With the limit, the internal dissipation falls slightly and the interior reaches 64.3 °C; the derated supply gives 3.57 W against a 3.40 W peak, a margin of 0.17 W. **R13 is met on paper** with the firmware port limit. Hosted sensors must tolerate the lower allowance on hot days. The voltage drop is not a problem: one port at 3 W over 10 m of 0.34 mm² cable loses 0.25 V (2.1 %), and the 1 m cable to the head, at 0.27 A, loses 0.027 V.

■ 12. Mass

Table 5. Mass estimate for the constructable design (LPN-DDR-003).

Assembly	Mass	Basis
Controller	0.33 kg	Dome from the model volume in ASA (0.077 kg), base at 35 % of its solid envelope in PC plus brass blades (0.092 kg), mains board, spacers, standoffs and screws (0.033 kg), estimates for the parts inside
Sensor head with clamps and 1 m cable	0.42 kg	Box and lid from the model volume in PC (0.156 kg); folded aluminium bracket, rubber strips and two band clamps (0.084 kg); printed internal plate at 60 % infill (0.018 kg); estimates for the rest

TRL 2 gave 0.25 kg for the controller; the concept model gave 0.34 kg and 0.45 kg. The constructable design is a little lighter because the solid bracket block of the concept became a folded 2 mm channel and the power supply envelope is a real 5 W module size. No requirement sets a mass.

■ 13. Cost (R16)

Value-engineering target: USD 150 (`budget_usd` , a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 143.00 (USD 7.00 under the target). `bom/bom.csv` has 18 lines, all priced: USD 75.00 for the controller (items 1 to 9 and 17), USD 61.00 for the sensor head, cable and ports (items 10 to 15 and 18) and USD 7.00 of hardware. Making the design constructable added line 17, the controller boards and fixings (USD 7.00), and line 18, the printed head plate (USD 4.00), and repriced line 15, the folded bracket (USD 6.00 to 7.00), and line 16, sealing washers and cable ties (USD 6.00 to 7.00); the total rose from USD 130.00. At 148 kWh a year the parts pay back in 3.9 years at USD 0.25 per kWh and 9.7 years at USD 0.10 per kWh, before installation labor.

■ 14. Results

Table 6. Requirement status at TRL 3.

ID	Requirement	Result	Status
R1	Fit the socket	Representative ANSI C136.41 base in the model; fit and a 5 min change cannot be checked on paper	Not verifiable at TRL 3
R2	Mains supply range	Supply 85 to 305 V; 385 V class varistor; scope 120 to 277 V (LPN-DDR-001 item 9)	Met (design review)
R3	Switch the luminaire	4.44 A steady at 400 W; inrush 72 A per 100 W driver, 289 A for four; about 20 A with zero-cross closing	At risk
R4	Dim the luminaire	2.44 mV steps; 10 mA sink serves 5 drivers	Met (design review)
R5	Schedule without the network	Crystal 160 s at -10 °C and 318 s at -30 °C in 30 days; TCXO clock 9.1 s	Met on paper with the TCXO clock
R6	Detect presence	Beam covers 7.5 to 47.9 m; 36 dB at 15 m; 0.63 s to full; rain, trees and cross-section unverified	At risk
R7	Save energy	36.4 % ideal driver, 34.1 % dimmed driver model, target 35 %	At risk
R8	Measure energy	1.65 % worst case with a one-point calibration (2.35 % without)	Met on paper, calibration required
R9	Report faults	0.144 J last gasp against 1.32 J stored	Met (design review)
R10	Low self-consumption	0.68 W average	Met on paper
R11	Survive the environment	Inside 63 to 65 °C at 45 °C in sun (70 °C rating); 118 J in the varistor at 5 kA; IP66 and surge need tests	At risk
R12	Fail safe	Normally closed relay; open 0 to 10 V line gives full output; coil drive needs a toggling signal; day needs both clock and light sensor	Met (design review)
R13	Host other sensors	0.25 V drop at 10 m; ports limited to 2.5 W above 50 °C inside: 3.40 W peak against 3.57 W at 64 °C (3.90 W against 3.55 W without the limit)	Met on paper (firmware port limit)
R14	Privacy	Doppler radar, no image; presence counts only	Met (design review)
R15	Secure and open	LoRaWAN 1.0.4, US915, published payload; 25.7 s a day at SF9, inside the 400 ms dwell limit; TALQ in CityTwin	Met (design review)
R16	Affordable	USD 143.00 against the USD 150 value-engineering target	Under the target by USD 7.00

Summary: 0 not met, 4 at risk, 11 met on paper or by design review, 1 not verifiable at TRL 3.

■ 15. Limits of this note

- No part is chosen. Every load, inrush, efficiency and derating figure is typical, and each must be replaced by datasheet values before TRL 4.
- The radar model says nothing about false triggers from trees, rain intensity or traffic. Only a field trial can settle R6.
- The thermal model is a single node with a fixed film coefficient; the luminaire top temperature is assumed.

- The energy results depend on the reference profile, which is an estimate only. LampNode ships with photocell-equivalent output (LPN-DDR-001 item 8); the road owner sets any dimming.

Safety: LampNode connects to mains voltage (up to 277 V AC nominal, 305 V maximum) and is installed at height beside traffic. The fail-on behavior in R12 depends on the coil drive dropping out when the controller stops, and on each driver model giving full output with an open 0 to 10 V line; both must be checked on real parts. Relay contacts welded by inrush would leave a lamp burning in daylight, not dark, but the fault must still be reported. Only qualified electricians may fit or remove the controller.